

DentaGuru System Documentation

Complete Technical Architecture & Operation Manual

DentaGuru Mobile & Web Application Documentation

Comprehensive Technical & Operating Manual

Version 2.0 | Google DeepMind Agentic Systems Architecture

1. Executive Summary & System Overview

DentaGuru is an end-to-end, multi-portal digital dental healthcare ecosystem built to streamline patient consultations, clinic management, specialist triage, and real-time electronic health records (E-Prescriptions).

Core Technological Stack

- **Frontend App:** Cross-platform Flutter Web & Mobile application (``lib/main.dart``).
- **Backend API:** Node.js & Express.js RESTful API Server operating on Port ``5000``.
- **Database Layer:** Supabase Cloud PostgreSQL Relational Database (``fommrwxpqzkctweosgp.supabase.co``).
- **State Management:** ``PatientProblemService`` singleton service with auto-syncing reactive listeners and local storage fallback.

2. Multi-Role Ecosystem & User Portals

DentaGuru provides tailored workspaces for 4 distinct user roles:

Role	Target Route	Primary Responsibilities
Patient	<code>`/patient`</code>	Submit oral health symptoms, view assigned specialists, track appointment status, access digital E-Prescriptions in Health Locker.
Dentist	<code>`/dentist`</code>	Receive forwarded symptom requests from Admin, accept & schedule appointments, issue live digital E-Prescriptions, log 3D tooth conditions.
Clinic Manager	<code>`/clinic`</code>	Manage clinic branch profile, configure procedure fee rate cards (Consultation, Root Canal, Orthodontics), monitor daily patient queue.
Super Admin	<code>`/admin`</code>	Triage incoming patient requests, assign specialists, register new doctors/clinics, remove users/doctors, monitor real-time practice revenue.

3. End-to-End Workflows & Data Pipelines

A. Patient Symptom Triage Pipeline

1. **Submission:** Patient fills in problem category, symptom description, and severity on `/patient`. Initial status: `Pending Admin Review`.
2. **Admin Triage:** Admin reviews the pending request on `/admin`, selects an available specialist, and clicks `Send Doctor Recommendation`. Status changes to `Pending Acceptance`.
3. **Doctor Acceptance:** The assigned Doctor receives the notification under **Admin Forwarded Symptoms** on `/dentist` and clicks `Accept & Schedule`. Status changes to `Accepted` (Green badge).
4. **Live Synchronization:** The appointment status updates live on the Patient's dashboard to `Confirmed`.

B. Live E-Prescription Delivery Pipeline

1. Doctor opens `/dentist` → clicks `Issue E-Prescription`.
2. Doctor specifies medicine name, dosage, duration, and instructions.
3. System saves the record to Supabase `medical\_records` table and updates local state.
4. Patient's Health Locker on `/patient` receives the digital prescription slip instantly.

C. Rates & Time Slot Management

1. Doctor/Clinic can update consultation fees and next available slots from `/admin`, `/clinic`, or directly from the Doctor greeting banner on `/dentist`.
2. Procedure rate cards (General Consultation, Tooth Decay, Root Canal, Orthodontics) update live across all doctor cards.

4. Supabase Database Schema

DentaGuru uses 6 relational tables in Supabase PostgreSQL:

1. **`users`:**
 - `id` (UUID), `name`, `email`, `phone`, `role` (`Patient`, `Dentist`, `Admin`, `Clinic`), `password` (BCrypt hash), `created\_at`.
2. **`dentists`:**
 - `id` (UUID), `user\_id` (FK), `clinic\_id` (FK), `speciality`, `license\_number`, `availability\_status`, `rating`, `reviews\_count`.
3. **`clinics`:**
 - `id` (UUID), `user\_id` (FK), `clinic\_name`, `location`, `verified`, `pricing` (JSONB array of procedure rates).
4. **`appointments`:**
 - `id` (UUID), `patient\_id` (FK), `dentist\_id` (FK), `clinic\_id` (FK), `date`, `time\_slot`, `treatment`, `status` (`pending`, `confirmed`, `completed`).

5. `medical\_records`:

- `id` (UUID), `patient\_id` (FK), `dentist\_id` (FK), `diagnosis`, `prescriptions` (JSONB array), `notes` (JSONB metadata).

6. `chat\_messages`:

- `id` (UUID), `sender\_id`, `receiver\_id`, `message\_text`, `created\_at`.
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5. API Endpoint Reference

- `POST /api/v1/auth/register`: Register new user (Patient, Dentist, Clinic, Admin).
 - `POST /api/v1/auth/login`: Authenticate user and return JWT tokens + role routing.
 - `GET /api/v1/appointments`: Fetch live appointments from Supabase.
 - `POST /api/v1/appointments`: Create/book new appointment.
 - `GET /api/v1/records`: Fetch medical records & E-Prescriptions for a patient.
 - `POST /api/v1/records`: Issue new digital E-Prescription.
 - `POST /api/v1/auth/reset-db`: Truncate all database tables for fresh testing.
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6. Workspace Quality Rules

1. **No Dummy Fallback Data:** Zero hardcoded dummy sample arrays in UI or controllers. Clean empty states when uninitialized.
2. **End-to-End Real-Time Persistence:** All user actions are immediately saved to Supabase PostgreSQL database tables.
3. **Auto-Sync on Launch:** Automatic sync calls (`syncAppointmentsFromApi()`, `fetchMedicalRecords()`, `fetchDentists()`, `fetchClinics()`) maintain 100% data fidelity on page refreshes.